

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT2-Constraint-Based Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS:3

Total Marks:30

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information. The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- Analyze all the given constraints and explain how they affect the robot's decision-making process.
- Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type
- Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Code of Conduct:

I V.Sireesha (192472208) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V.Sireesha

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6
Solution Development	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5

Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6
Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5

1. Hospital Shift Assignment using Backtracking Search

Problem Understanding

The problem is to assign 3 doctors (D1, D2, D3) to 3 different shifts (Morning, Afternoon, Night) using the Backtracking Search Algorithm. The objective is to find a valid schedule while satisfying all the given constraints.

Constraints

1. D1 cannot work the Night shift.
2. D2 must be assigned before D3 (Morning < Afternoon < Night).
3. Only one doctor can be assigned to each shift.
4. D3 cannot work the Morning shift.
5. Every doctor must be assigned exactly one shift.

Constraint Analysis

Constraint	Analysis
D1 cannot work Night →	D1 can only be assigned Morning or Afternoon.
D2 before D3 →	The shift assigned to D2 must occur earlier than D3's shift.
One doctor per shift →	No two doctors can share the same shift.
D3 cannot work Morning →	D3 can only work Afternoon or Night.
One shift per doctor →	Every doctor must be assigned one unique shift.

Priority of Constraints

1. Eliminate invalid shifts (D1 ≠ Night, D3 ≠ Morning).
2. Maintain one doctor per shift.
3. Ensure D2 is scheduled before D3.
4. Assign all doctors successfully.

Solution Development using Backtracking Search

Step 1: Assign D1

Possible shifts:

- Morning
- Afternoon

Choose Morning.

Doctor	Shift
D1	Morning

Constraint Check:

- D1 is not Night
- No duplicate shifts

Step 2: Assign D2

Available shifts:

- Afternoon
- Night

Try Afternoon.

Doctor	Shift
D1	Morning
D2	Afternoon

Constraint Check:

- Unique shift
- D2 is before D3 (will verify after D3 assignment)

Step 3: Assign D3

Remaining shift:

- Night

Check constraints:

- D3 is not Morning
- D2 (Afternoon) is before D3 (Night)
- One doctor per shift
- Every doctor assigned exactly one shift

All constraints are satisfied.

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

Constraint Checking

Constraint	Status
D1 cannot work Night	Satisfied
D2 before D3	Satisfied
One doctor per shift	Satisfied
D3 cannot work Morning	Satisfied
Every doctor assigned one shift	Satisfied

Application of Concepts

Search Technique: Backtracking Search

Backtracking is an AI search algorithm that builds a solution step by step. After each assignment, it checks whether all constraints are satisfied. If any constraint is violated, the algorithm backtracks to the previous step and tries another possible assignment. This reduces unnecessary searches and guarantees a valid solution if one exists.

Justification

Backtracking Search is the most suitable algorithm because it systematically explores all possible assignments while checking constraints at every step. Invalid assignments are rejected immediately, reducing the search space and improving efficiency. The final schedule satisfies every constraint without conflicts, making it the optimal solution.

2. Robot Path Finding using Breadth-First Search (BFS)

Problem Understanding

A robot moves in a 5×5 grid from the Start (S) position to the Goal (G) position. The robot must find the shortest path while avoiding obstacles using the Breadth-First Search (BFS) algorithm.

Constraints

1. The robot can move only Up, Down, Left, and Right.
2. Each movement has a cost of 1.
3. The robot cannot move through obstacles.
4. Manhattan Distance is used as the heuristic.
5. The robot must determine the optimal path and total cost.

Constraint Analysis

Constraint	Analysis
Up, Down, Left, Right movement	Robot explores only four possible directions.
Cost = 1	Every move increases the path cost by one.
Obstacles	Blocked cells cannot be visited.
Manhattan Distance	Used to estimate the distance to the goal.
Optimal Path	BFS guarantees the shortest path in an unweighted grid.

Solution Development using BFS

Grid Representation

S	0	X	0	0
0	0	X	0	0
0	0	0	0	X
X	X	0	X	0
0	0	0	0	G

Legend:

- S = Start
- G = Goal
- X = Obstacle
- 0 = Free cell

Step-by-Step BFS Node Expansion

Initial Queue

Queue = [S]
Visited = {S}

Step 1

Expand S
Queue : [(2,1), (1,2)]
Visited : S, (2,1), (1,2)

Step 2

Expand (2,1)
Add : (3,1), (2,2)
Queue : [(1,2), (3,1), (2,2)]

Step 3

Expand (1,2)

Right side blocked.

Queue : [(3,1), (2,2)]

Step 4

Expand (3,1)

Add : (3,2)

Queue : [(2,2), (3,2)]

Step 5

Expand (2,2)

Add : (3,2)

Already visited.

Queue : [(3,2)]

Step 6

Continue BFS until Goal.

Optimal path obtained : $S \rightarrow (2,1) \rightarrow (3,1) \rightarrow (3,2) \rightarrow (3,3) \rightarrow (3,4) \rightarrow (4,5) \rightarrow G$

Queue Updates

Step	Expanded Node	Queue After Expansion
1	S	(2,1),(1,2)
2	(2,1)	(1,2),(3,1),(2,2)
3	(1,2)	(3,1),(2,2)
4	(3,1)	(2,2),(3,2)
5	(2,2)	(3,2)
6	Continue	Goal Reached

Manhattan Distance Calculation

Formula: $[h(n)=|x_1-x_2|+|y_1-y_2|]$

Example: Start = (1,1) , Goal = (5,5)

$$= |5-1|+|5-1|$$

$$= 4+4$$

$$= 8$$

Manhattan Distance = 8

Application of Concepts

Search Technique

Breadth-First Search (BFS) explores the grid level by level. It visits every nearest node before moving to the next level. Since every move has the same cost (1), BFS guarantees the shortest path from Start to Goal while avoiding obstacles.

AI Concepts Applied

- State Space Search
- Breadth-First Search Algorithm
- Queue Data Structure
- Path Cost Calculation
- Manhattan Distance Heuristic

Justification

BFS is the most suitable algorithm because all moves have equal cost. It explores nodes systematically, avoids revisiting explored cells, and guarantees the shortest path to the goal. Manhattan Distance helps estimate how far the goal is from the current position, improving understanding of the search process.

3. Autonomous Rescue Robot using Online Search Agent

Problem Understanding

An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot starts at S (Start) and must reach G (Goal) while avoiding obstacles and minimizing the total travel cost. Since the environment is partially observable, the robot can only detect adjacent cells and must update its knowledge as it moves.

Given Constraints

1. The robot can move Up, Down, Left, and Right.
2. Some cells are blocked (obstacles) and cannot be crossed.
3. The robot has limited sensing capability and can observe only adjacent cells.
4. Each movement has a cost of 1, while entering a risky zone adds an extra cost of 2.
5. The robot must minimize the total cost and ensure safe navigation.
6. The robot must update its knowledge dynamically (Online Search scenario).

Constraint Analysis

Constraint	Analysis
Up,Down,Left,Right movement	The robot can move only in four directions.
Obstacles	Blocked cells cannot be entered and must be avoided.

Constraint	Analysis
Limited sensing	The robot knows only nearby cells and discovers the environment while moving.
Movement cost	Every step increases the path cost by 1.
Risky zones	Entering risky areas increases the total cost by 2.
Dynamic updates	The robot continuously updates its map and changes its path if needed.

Priority of Constraints

1. Avoid obstacles.
2. Minimize travel cost.
3. Avoid risky zones whenever possible.
4. Reach the survivor safely.
5. Update the path dynamically as new information is discovered.

Solution Development

Chosen Search Strategy: Online Search Agent (Uniform Cost Search can also be used)

Step 1 : The robot starts from S and observes the adjacent cells.

Step 2 : Safe neighbouring cells are added to the search list, while blocked cells are ignored.

Step 3 : The robot selects the path with the lowest total cost.

Step 4 : Whenever a new obstacle or risky zone is detected, the robot updates its map and recalculates the best path.

Step 5 : The process continues until the robot reaches the survivor (G).

Application of Concepts

a) Intelligent Agent : The rescue robot acts as an intelligent agent because it senses the environment, processes information, makes decisions, and performs actions to reach the survivor safely.

b) Rationality : The robot behaves rationally by choosing the safest and lowest-cost path while avoiding unnecessary risks and obstacles.

c) Environment Type

Environment Property	Description
Partially Observable	The robot can observe only nearby cells.
Dynamic	New obstacles or risky zones may appear.
Sequential	Every action affects future decisions.

Environment Property	Description
Deterministic	The robot's movement produces predictable results.
Discrete	The environment consists of grid cells.

Justification

The Online Search Agent is the most suitable approach because the robot does not have complete knowledge of the environment initially. It continuously gathers information, updates its knowledge, avoids obstacles, and selects the lowest-cost path. This approach ensures safe navigation, minimizes travel cost, and enables the robot to adapt to changes in the environment.